# Technical appendix / image controls and provenance

Original geometry, registered cameras, visual review and current limitations / supporting material, separate from the public design story.

## Sources and geometry

V9 complete-city geometry and ac96714 requirements snapshot. V10 adds four explicit open shade structures and two pedestrian links; no homes, jobs, school places or compute capacity are added. Zero-area clipping debris is removed from scene inputs. The local scene master polygon defines rendering extent only; it does not replace the design area or whole-site boundary.

## Scene conventions

Source Copernicus terrain is retained. Concept building platforms use maximum sampled corner height to avoid terrain clipping; foundations and grading require civil design. Procedural planting/PV/entourage are disabled. PV rows follow the explicit V9 assets. AI materials and people are illustrative. No long residential heat pipe is credited.

## Image generation

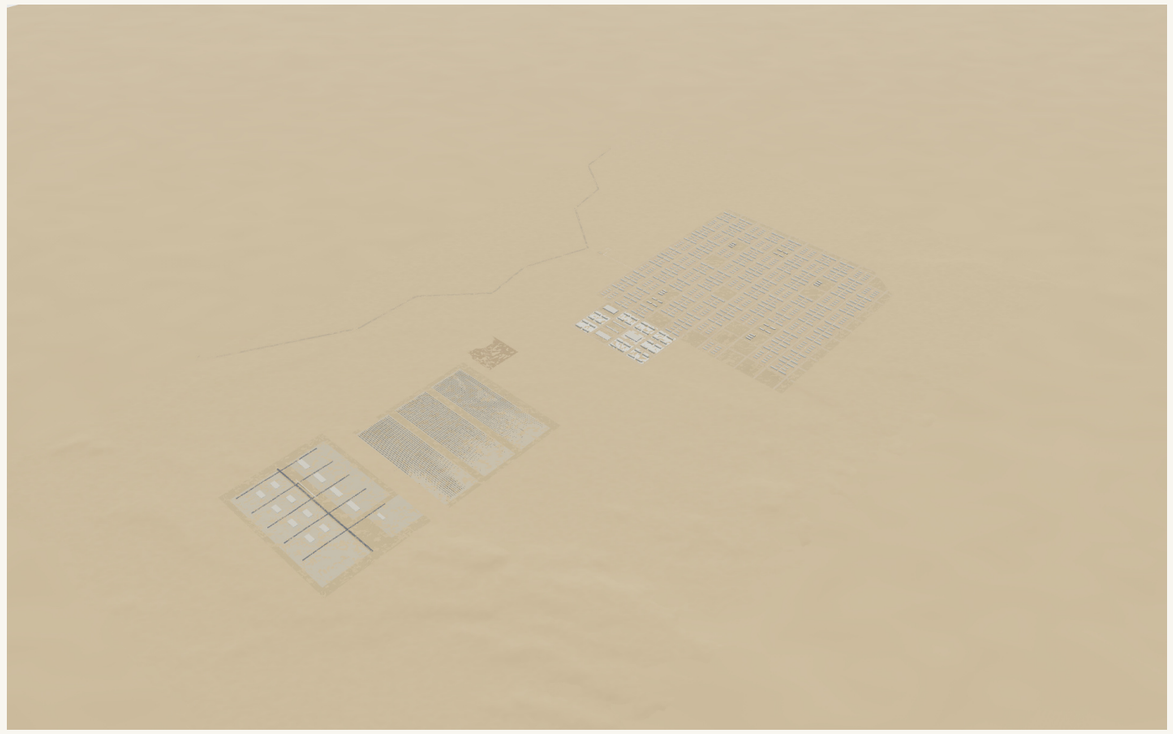
Built-in image_gen.imagegen; exact model version is not exposed. Full prompts and image/control hashes are saved in image_provenance.json and generation-record.json. Rejected trials were excluded: an eight-building simplification of P, oversized aerial people and an additional local-city trial that closed separate bar footprints into courtyards.

## Review scope

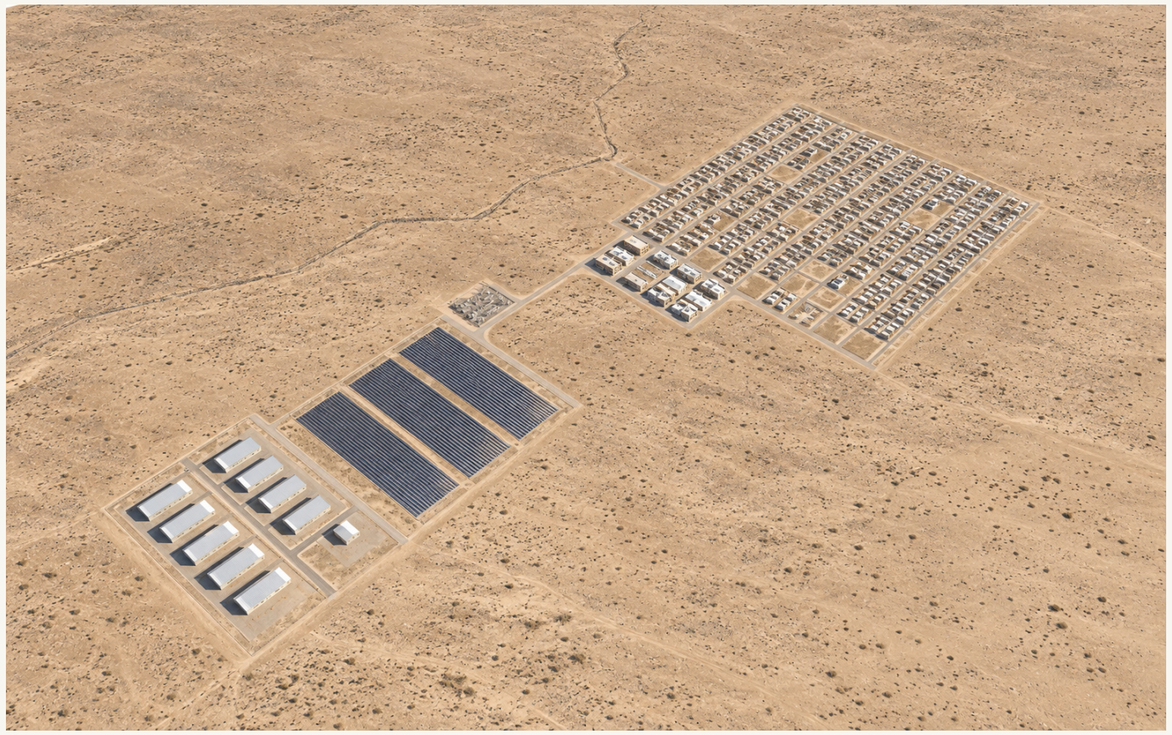
The selected images were visually compared for view direction, principal massing, road relationships and prohibited additions. This is a human-readable visual QA record, not a pixel-perfect geometric certification. Plans and underlying geometry govern dimensions and capacity. Rights, hydraulic levels and climate operation remain unverified.

# city-overall / control and finished illustration

Southwestern built settlement; not full competition territory / Hypothetical full build-out



**GEOMETRY-DERIVED CONTROL**



**SELECTED FINISHED CONCEPT IMAGE**

Position / UTM19S: [579300, 6550100, 6200]

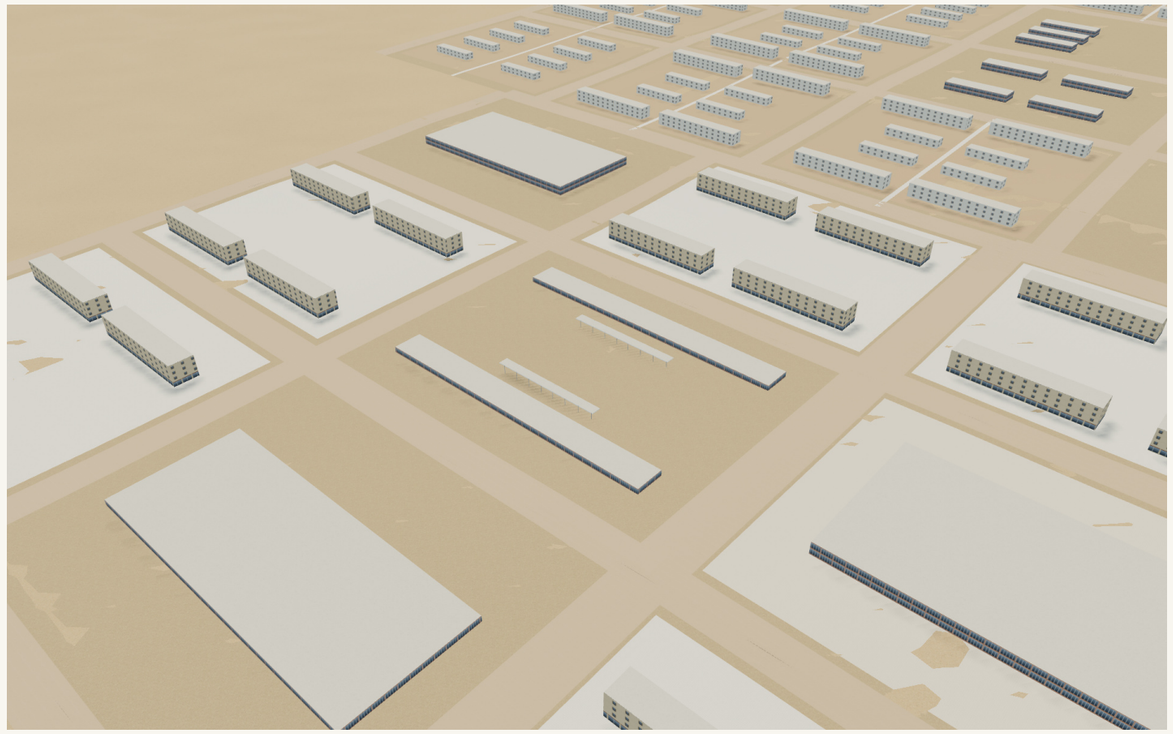
Target / UTM19S: [574300, 6555500, 0]

Perspective camera / vertical FOV 42 degrees / 1920 x 1200 control

The output preserves the intended spatial organisation. Finishes, surface textures, minor equipment and human activity are illustrative. No people, texture or planted ground establishes capacity or surveyed conditions. See the registry and source/control/image fingerprints for traceability.

# A03-aerial / control and finished illustration

Node camera crop / Phase 1 node core with full-build-out urban context



**GEOMETRY-DERIVED CONTROL**



**SELECTED FINISHED CONCEPT IMAGE**

Position / UTM19S: [574360, 6555880, 300]

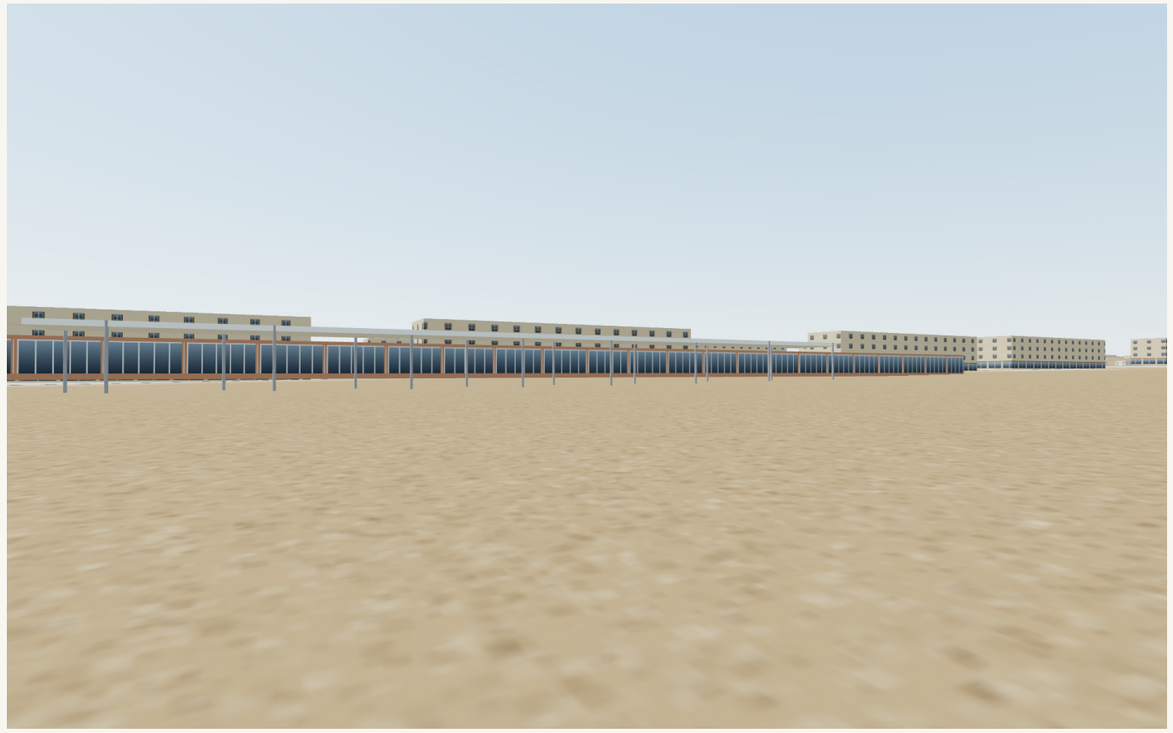
Target / UTM19S: [574075, 6556200, 0]

Perspective camera / vertical FOV 43 degrees / 1920 x 1200 control

The output preserves the intended spatial organisation. Finishes, surface textures, minor equipment and human activity are illustrative. No people, texture or planted ground establishes capacity or surveyed conditions. See the registry and source/control/image fingerprints for traceability.

# A03-eye / control and finished illustration

Node camera crop / Phase 1 node core with full-build-out urban context



**GEOMETRY-DERIVED CONTROL**



**SELECTED FINISHED CONCEPT IMAGE**

Position / UTM19S: [574054, 6556181, 1.6]

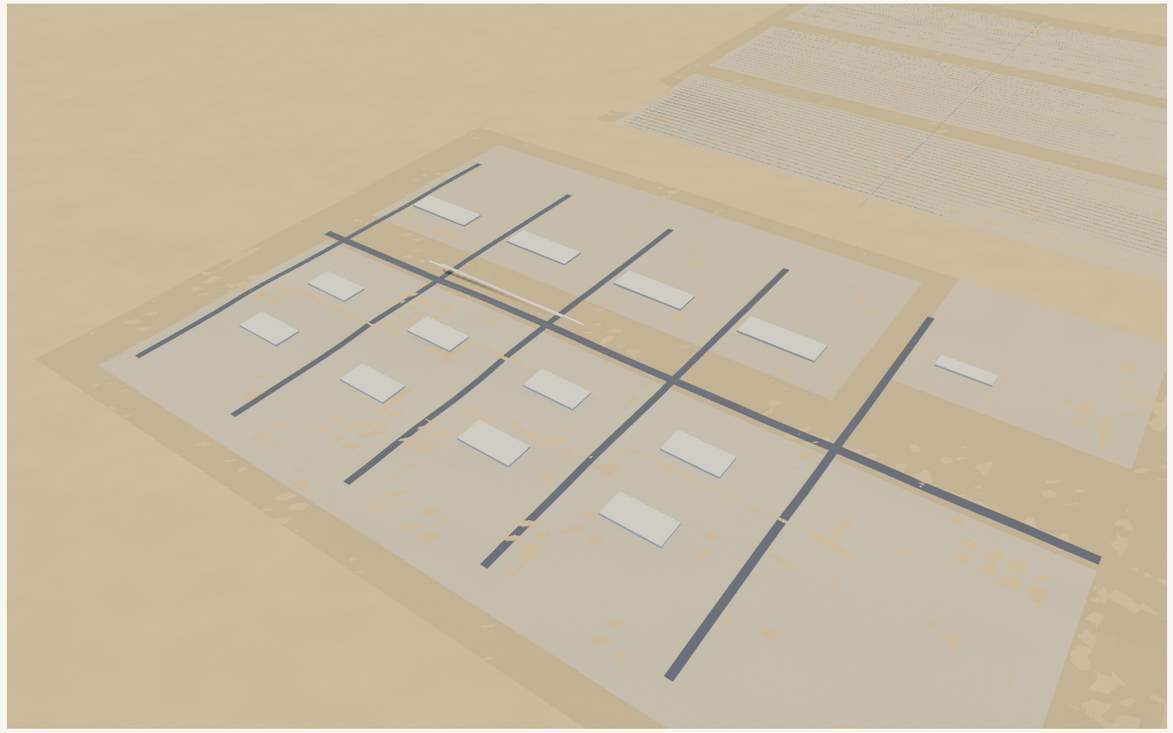
Target / UTM19S: [574088, 6556248, 2]

Perspective camera / vertical FOV 62 degrees / 1920 x 1200 control

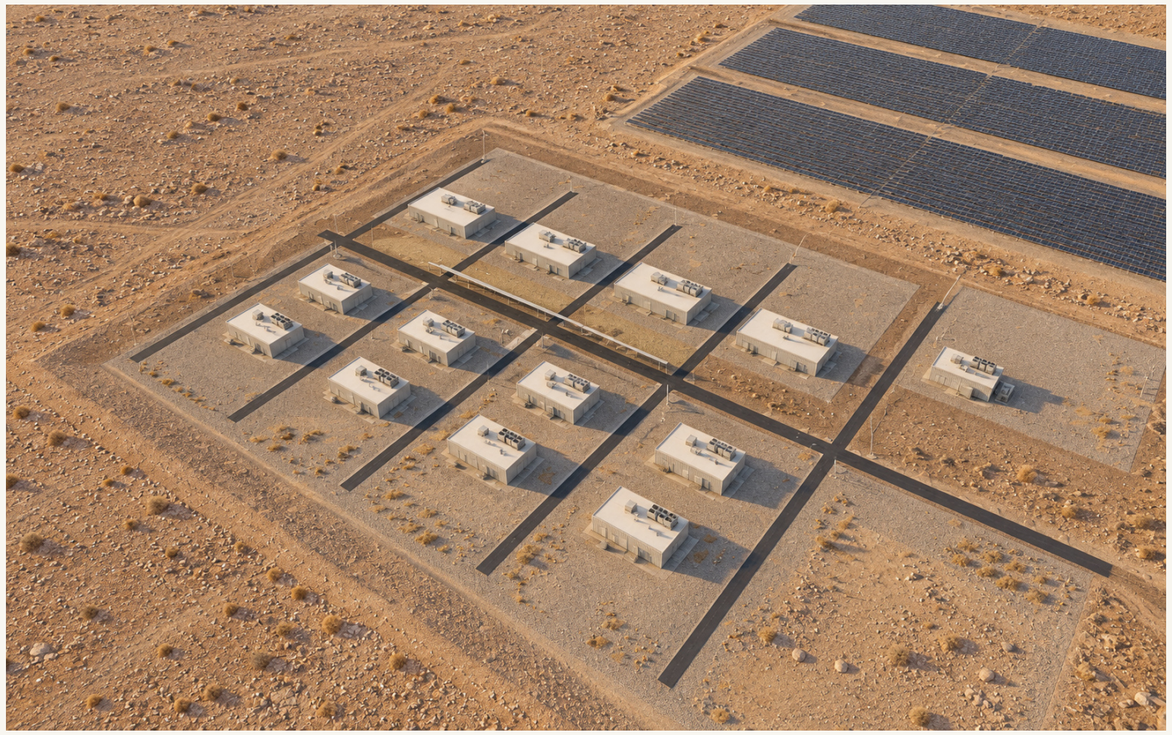
The output preserves the intended spatial organisation. Finishes, surface textures, minor equipment and human activity are illustrative. No people, texture or planted ground establishes capacity or surveyed conditions. See the registry and source/control/image fingerprints for traceability.

# B03-aerial / control and finished illustration

Node camera crop / P1-P3 full build-out; halls are staged 2 / 4 / 8 MW



**GEOMETRY-DERIVED CONTROL**



**SELECTED FINISHED CONCEPT IMAGE**

Position / UTM19S: [574500, 6551600, 900]

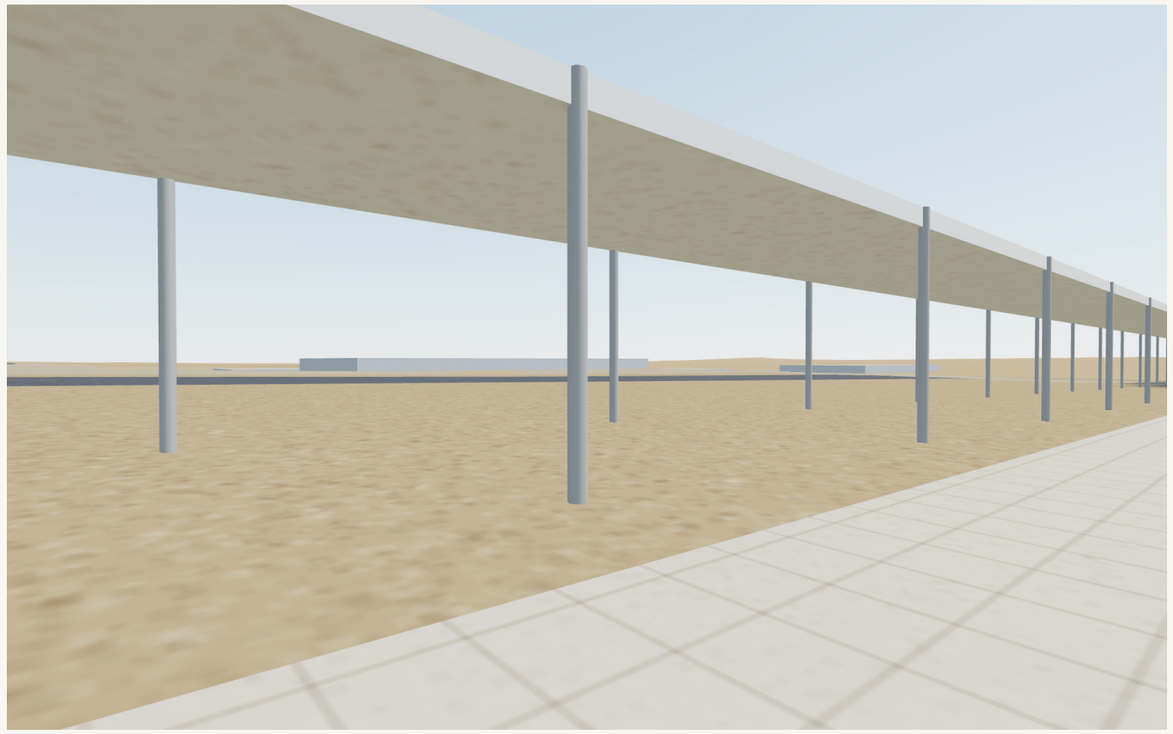
Target / UTM19S: [573850, 6552500, 0]

Perspective camera / vertical FOV 44 degrees / 1920 x 1200 control

The output preserves the intended spatial organisation. Finishes, surface textures, minor equipment and human activity are illustrative. No people, texture or planted ground establishes capacity or surveyed conditions. See the registry and source/control/image fingerprints for traceability.

# B03-eye / control and finished illustration

Node camera crop / P1-P3 full build-out; halls are staged 2 / 4 / 8 MW



**GEOMETRY-DERIVED CONTROL**



**SELECTED FINISHED CONCEPT IMAGE**

Position / UTM19S: [573695, 6552589, 1.6]

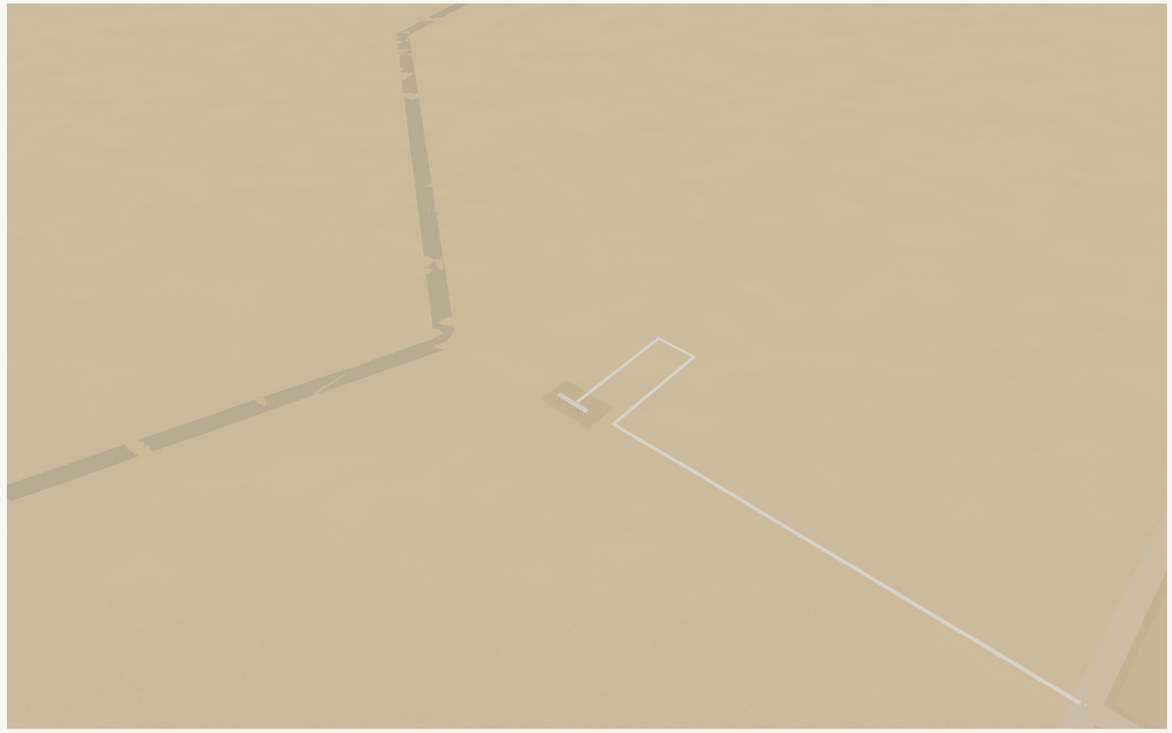
Target / UTM19S: [573570, 6552440, 3]

Perspective camera / vertical FOV 62 degrees / 1920 x 1200 control

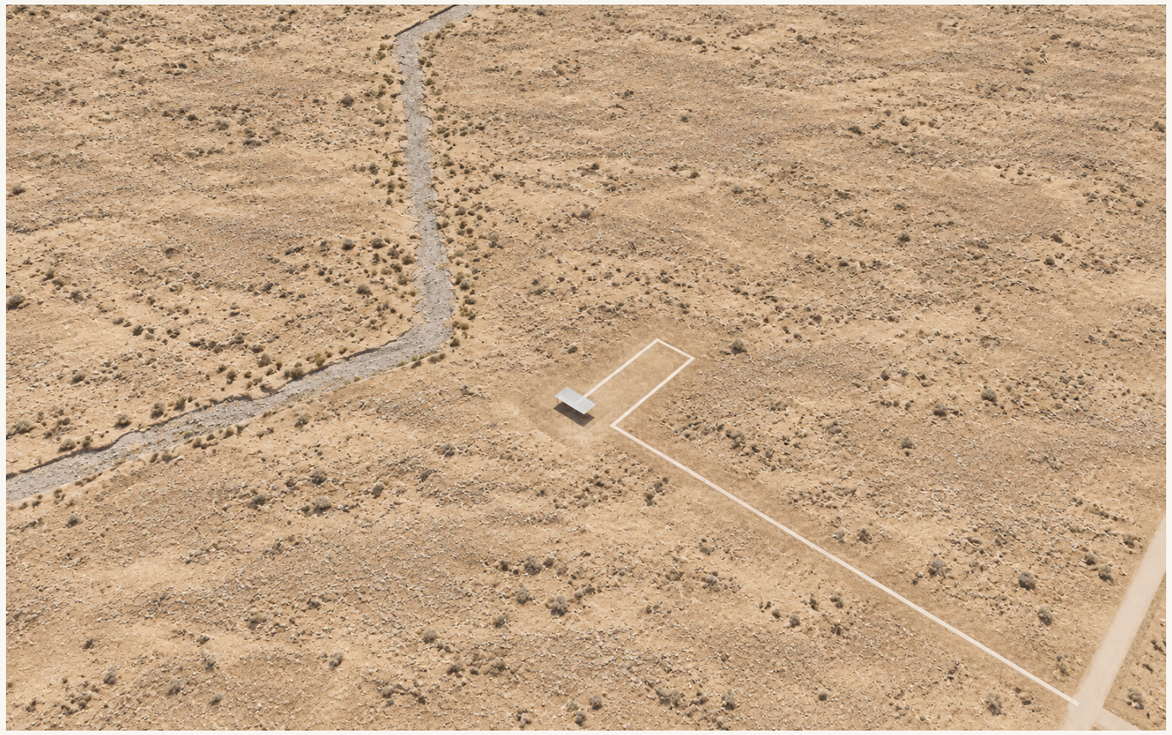
The output preserves the intended spatial organisation. Finishes, surface textures, minor equipment and human activity are illustrative. No people, texture or planted ground establishes capacity or surveyed conditions. See the registry and source/control/image fingerprints for traceability.

# C03-aerial / control and finished illustration

Node camera crop / Phase 2



**GEOMETRY-DERIVED CONTROL**



**SELECTED FINISHED CONCEPT IMAGE**

Position / UTM19S: [573440, 6556650, 650]

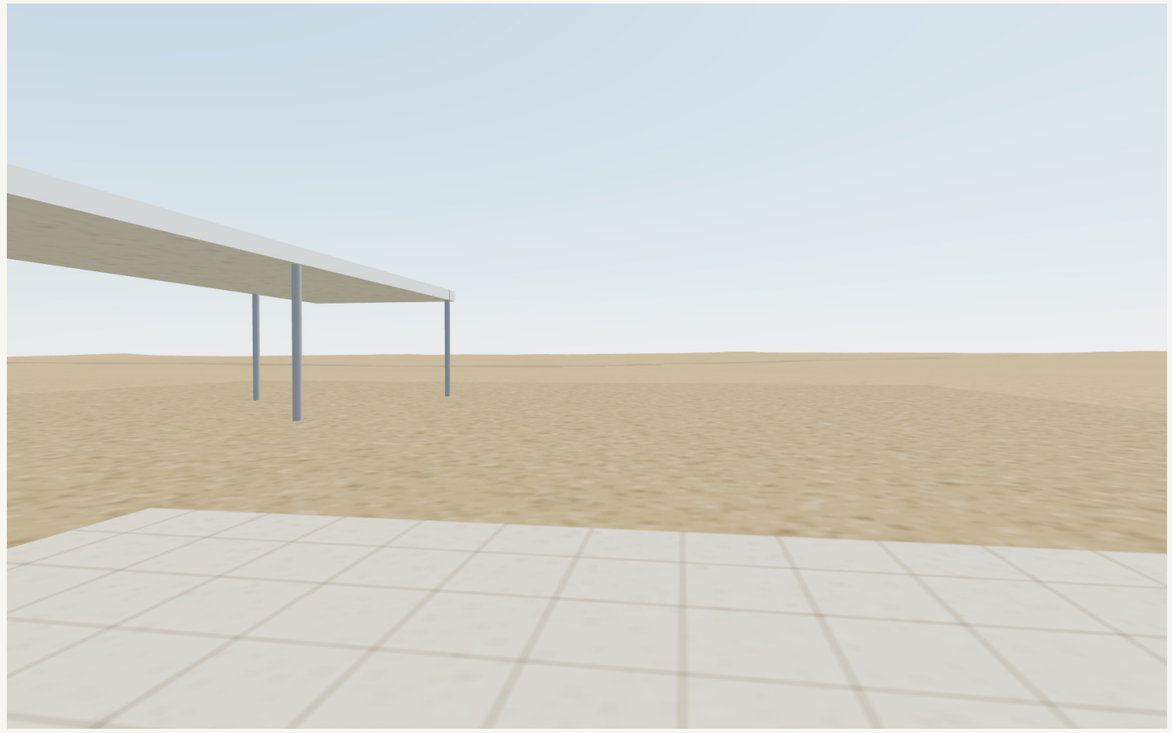
Target / UTM19S: [572950, 6557210, 0]

Perspective camera / vertical FOV 44 degrees / 1920 x 1200 control

The output preserves the intended spatial organisation. Finishes, surface textures, minor equipment and human activity are illustrative. No people, texture or planted ground establishes capacity or surveyed conditions. See the registry and source/control/image fingerprints for traceability.

# C03-eye / control and finished illustration

Node camera crop / Phase 2



**GEOMETRY-DERIVED CONTROL**



**SELECTED FINISHED CONCEPT IMAGE**

Position / UTM19S: [572983, 6557165, 1.6]

Target / UTM19S: [572795, 6557135, 0]

Perspective camera / vertical FOV 62 degrees / 1920 x 1200 control

The output preserves the intended spatial organisation. Finishes, surface textures, minor equipment and human activity are illustrative. No people, texture or planted ground establishes capacity or surveyed conditions. See the registry and source/control/image fingerprints for traceability.

# Evidence and remaining verification

Internal geometry checks and declared scenarios are not organiser design approval.

Geometry and accounts
V9 has 815 source building footprints, 91 residential blocks and 10240 dwelling capacity. Building overlap: 0.0 m2. Buffered street reserve conflicts: 0. All district land accounts close. Four V10 open canopies have no household or employment capacity. V9 geometric checks do not certify the V10 yard-spur crossing details; these remain an explicit design hold-point.

Capacity assumptions
2.5 residents/dwelling; 120 m2 gross housing floor area/dwelling; education need 15% of residents; 360 places per school wing. Job totals are programme capacity, not contracted jobs. Roster beds are counted separately. Phase-one combined existing/new water rights and fixed-pilot demand remain unresolved.

Reproduction and disclosure
Planning: tools/complete_city_v9.py. Scene and controls: tools/presentation_v10.py; controls rendered with the project renderer plus the disclosed local platform/PV orientation adjustment in renderer-v10.js.
Presentation: tools/publish_v10.py. Human directed whole-site scope, regional legibility, housing continuity, finished rendering and close framing; Codex developed the design/geometry and composed the presentation.
No public website update or new automatic 12/12 approval is claimed by this presentation review.